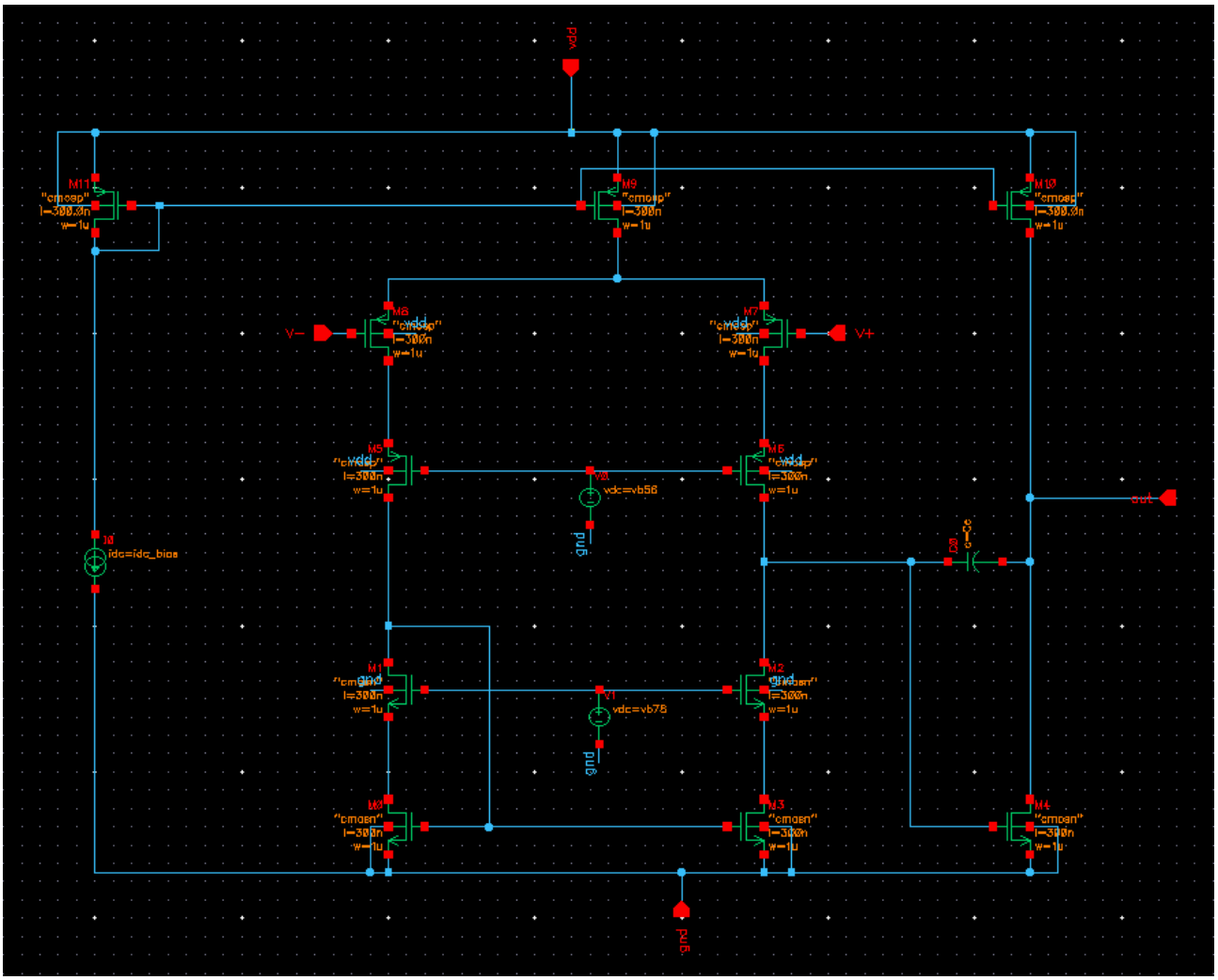
# EE20B003 Assignment 7

Anurag Manoj, EE22B003

The following op amp with pmos input pair has been designed and tested-




## Specifications

| | |
|---|---|
| Closed Loop gain | 9 |
| Closed Loop Bandwidth Required | 17 MHz |
| Load Capacitor CL | 8 pF |

| | |
|---|---|
| Closed Loop gain | 9 |
| Load Resistor | 2.3 kOhms |
| Input Resistance | 13 kOhms |

**Component Values**

| | |
|---|---|
| All MOSFET Lengths | 300 nm |
| Width of all mosfets | 1 um |
| Multiplier (m00) | 2 |
| VDD | 1.8V |
| Vb56 | 560m |
| Vb78 | 870m |
| Ad=As | 0.36 (u)^2 |
| Pd=Ps | 2.72u |
| Bias Current | 16u |
| pmos tail current source multiplier | 32 |
| pmos input pair multiplier | 47 |
| pmos cascode mosfet multiplier | 50 |
| nmos common gate cascode mosfet pair multiplier | 9.4 |
| nmos cascode mirror (bottom nmos) multiplier | 16 |
| nmos 2nd stage multiplier | 75 |
| pmos 2nd stage current source multiplier | 150 |

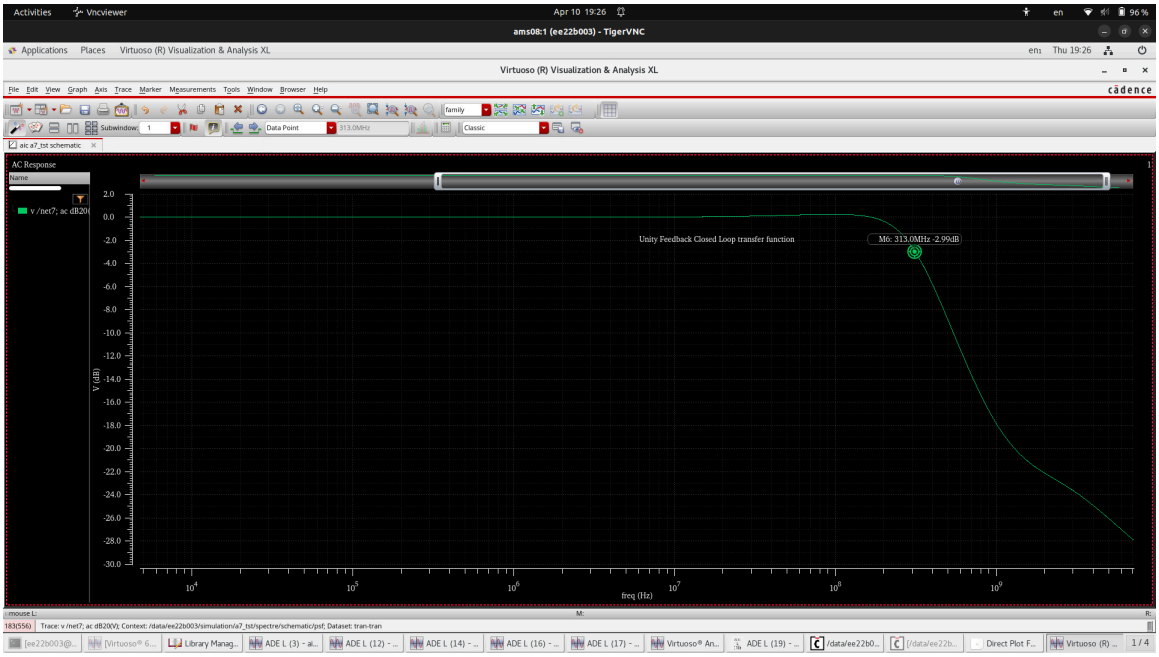
**Simulation Result**

| | |
|---|---|
| Power | 1.523 mW |
| HD3 | -23.16 dB |
| Closed Loop Bandwidth (3dB) | 21.9 MHz |
| Closed loop DC gain | 8.956 |
| Unity Loop Gain Frequency | 13.2 MHz |
| Op-amp Open Loop gain | 18852 |
| Phase Margin of amplifier | 59.65 degrees |
| Slew Rates (+/-) | 26.35/65.6 V/us |
| Swing Limits | 850mV to 1V i/p, 10mV to 1.31V o/p with load resistor, 800mV to 1V i/p and 20mV to 1.78V o/p with no load resistor |

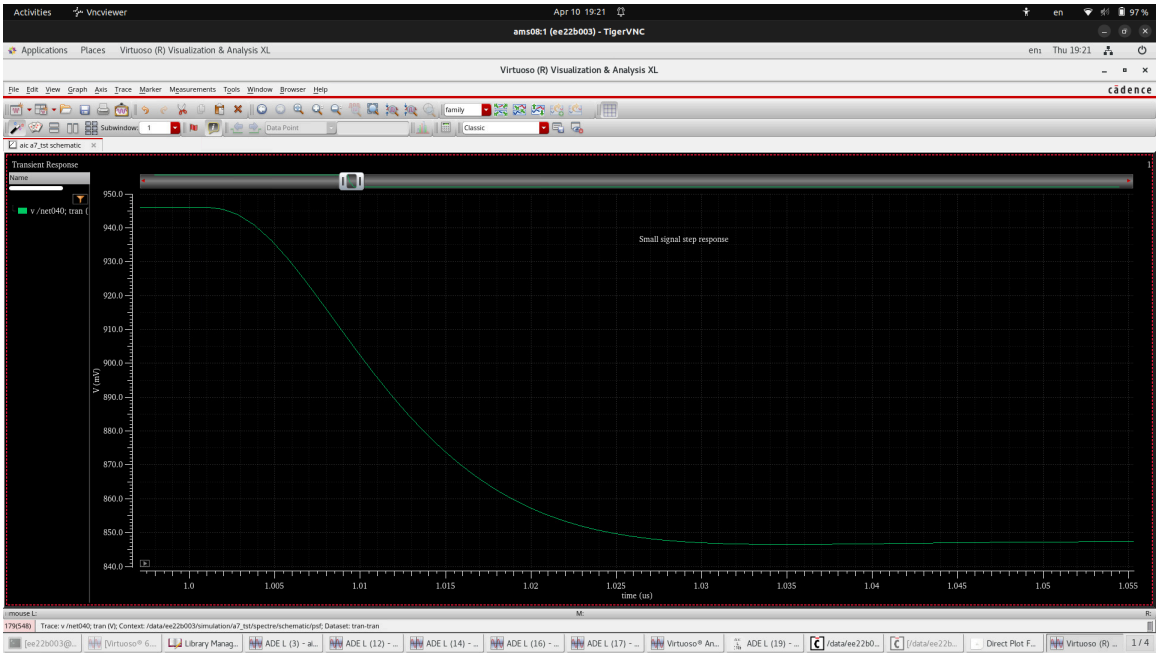
**Noise Simulation Results**

| | |
|---|---|
| Noise fraction due to RL | 0.01% |
| Noise fraction due to Ri | 63.72% |
| Noise fraction due to Rf | 7.07% |
| Noise fraction due to 1st stage | 22.22% |
| Noise fraction due to 2nd stage | 0.26% |
| Integrated output noise | 180uV |

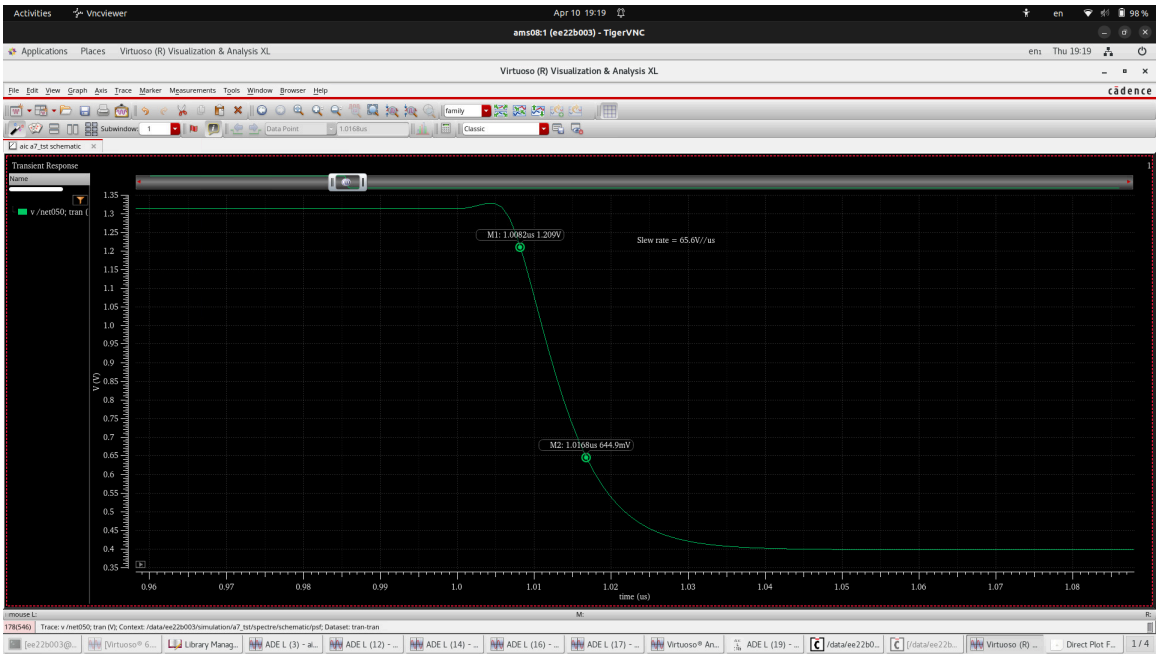
Unity Gain Closed Loop Transfer function



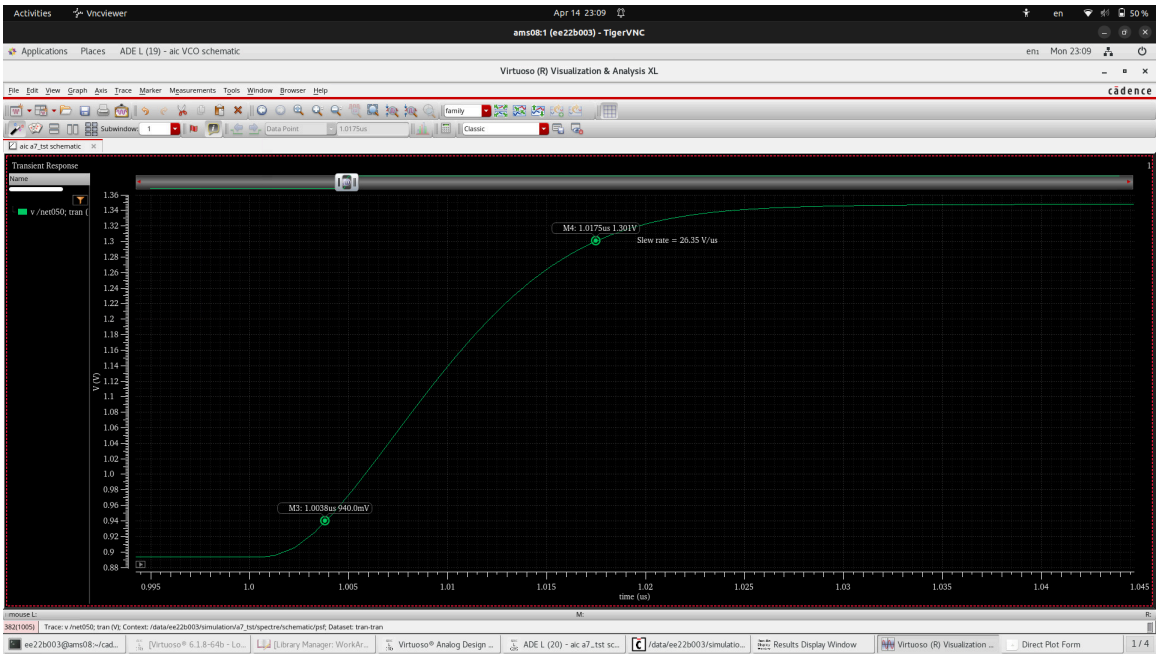

Small Signal Step Response



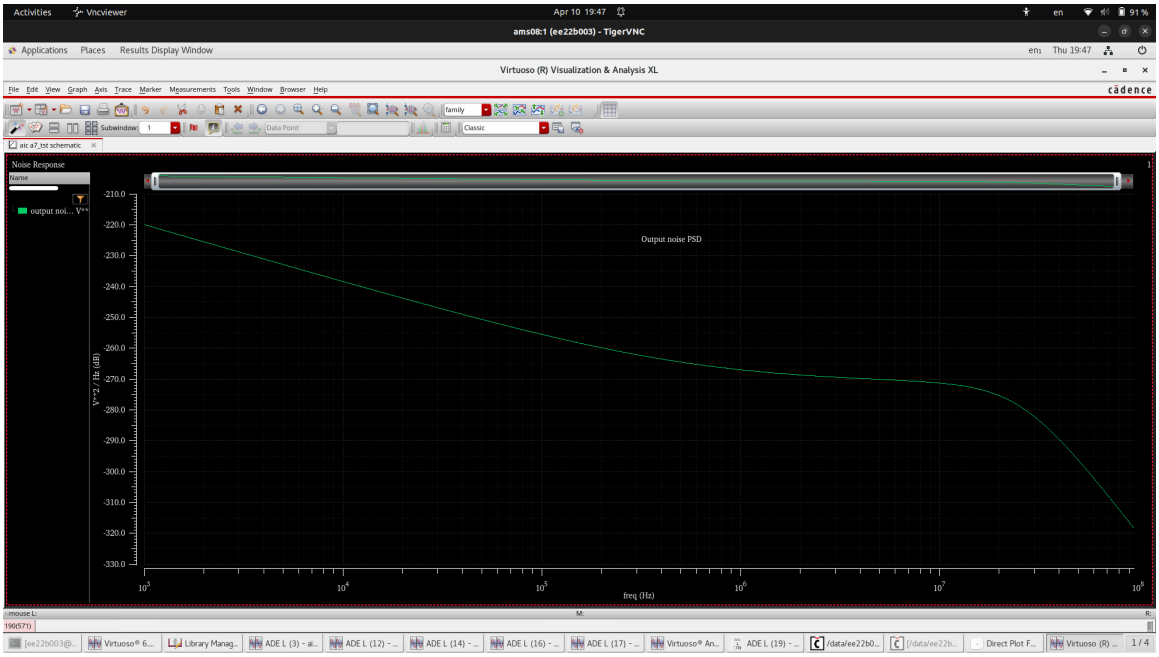

Slewing_(-ve)_(Large signal step response)



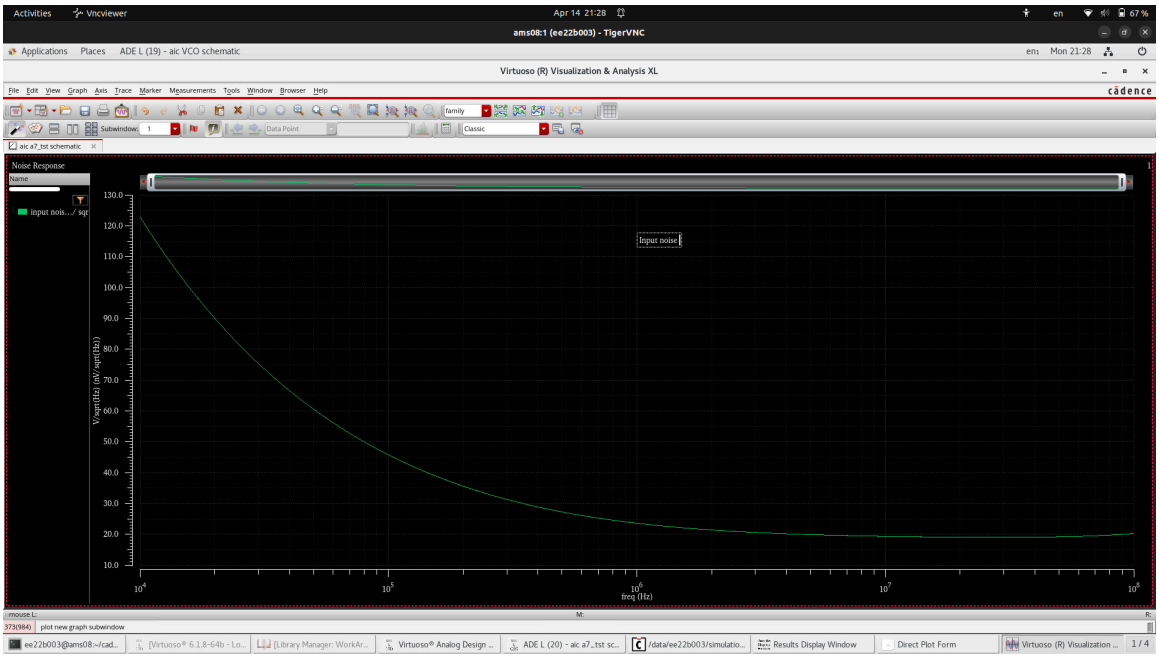

Slewing_(+ve)_(Large signal step response)



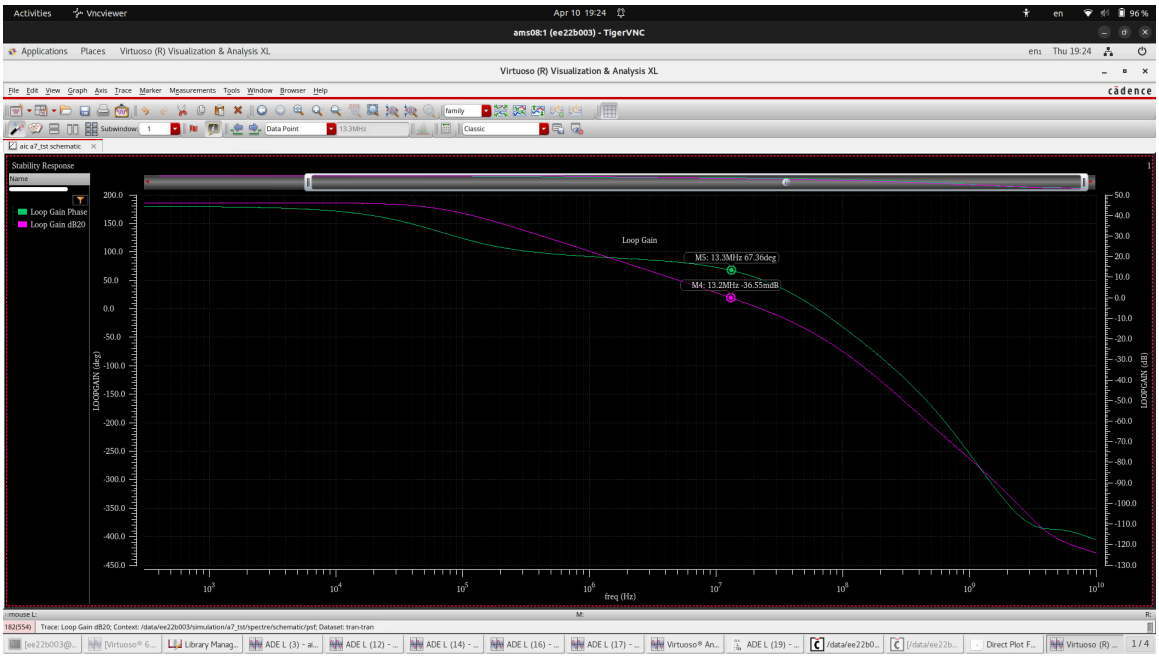

## Output Noise PSD



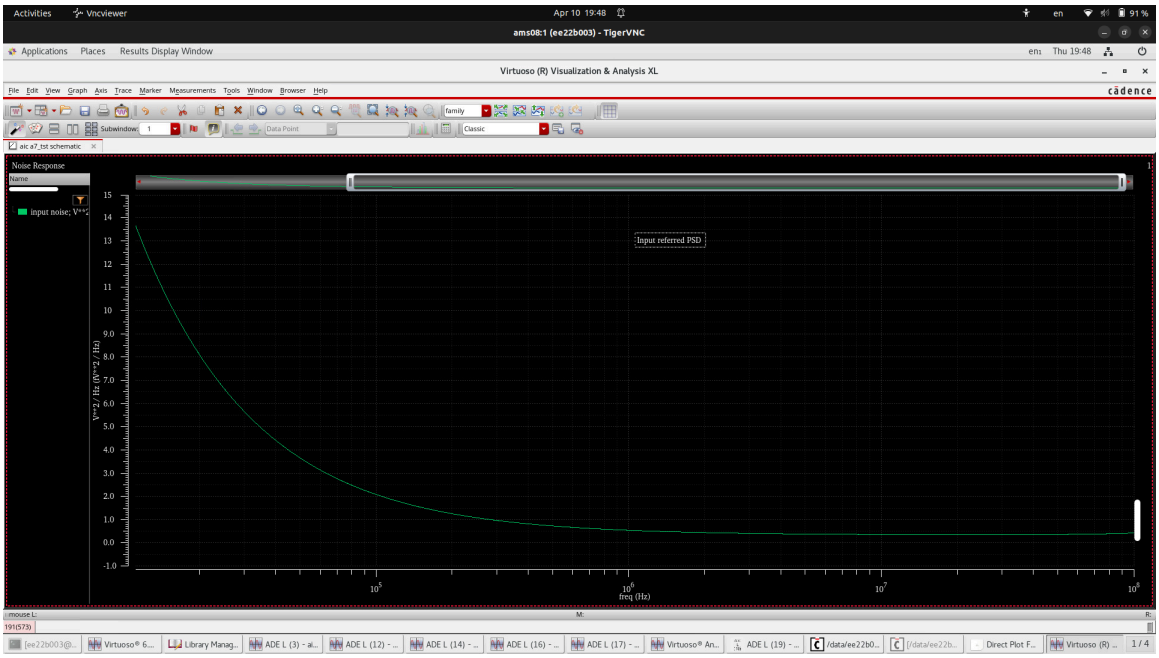

## Input referred noise PSD



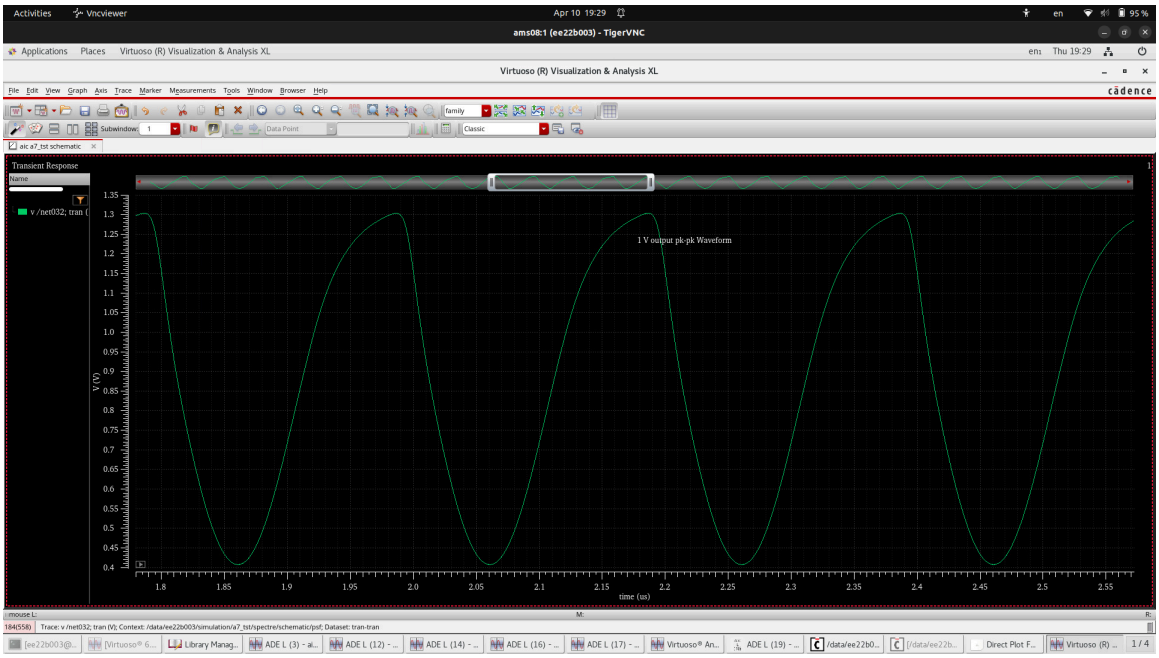

## Loop Gain Phase and Magnitude Bode Plot




## Input Referred PSD




1V peak to peak output for large signal sin wave input



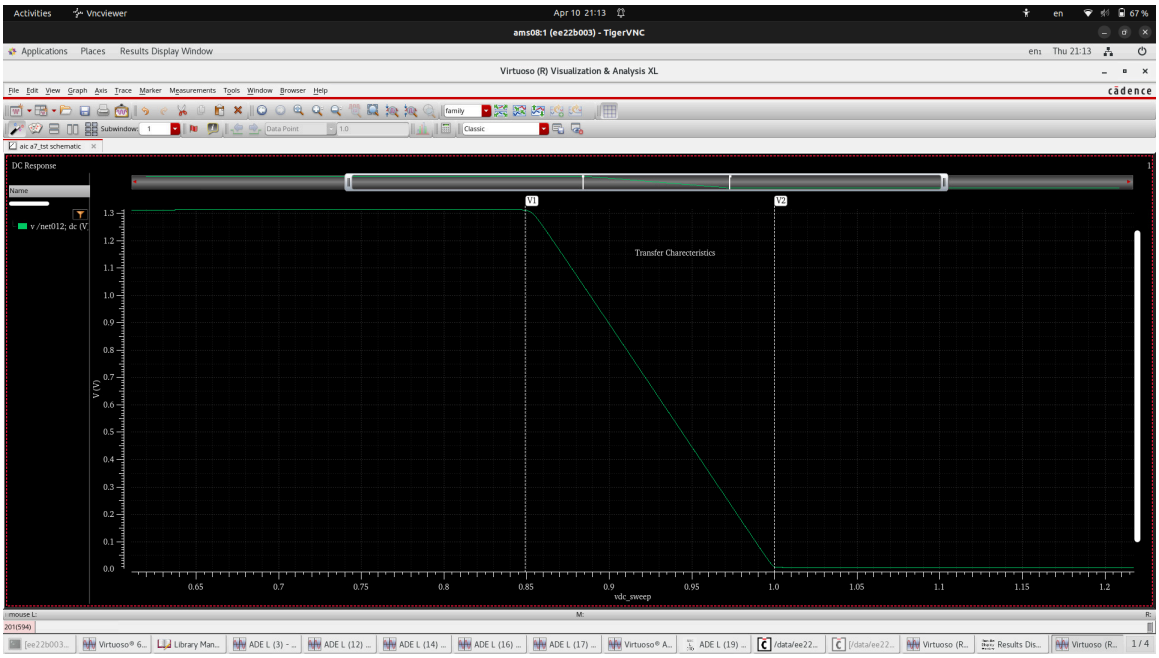

Closed Loop Transfer Function




## DC transfer characteristics without no load resistor




## DC transfer characteristics with load resistor




## Design Procedure

From MOS Characterization, the current and gm per 1um/0.3um of transistor is -

| | | |
|---|---|---|
| VDSAT | 0.15V | 0.25V |
| PMOS | 2.5uA, 27uA/V | 8uA, 48 uA/V |
| NMOS | 10.8uA, 113.5uA/V | 39uA, 200 uA/V |

Also, for pmos and nmos, the threshold voltage is empirically related to body effect as

PMOS: Vth = 0.2591 * Vb + 0.4593  (V/V and V)

NMOS: Vth = 0.2470 * Vb + 0.4707  (V/V and V)

Now considering vgs~vdsat+vth, and giving 50mV margins for each mosfet source, as well as fixing vdsats, all the operating points and bias voltages can be determined, as well as the ratio of mosfet sizes, in first and second stage. Then the size of nmos in 1st stage is tweaked, allowing the bottom most nmos to have a larger vdsat, and a smaller gds, as the nmos cascode limits the Rout and hence gain of first stage. Similarly, an nmos width larger than pmos width is chosen for second stage, to maximize gm for a given Ids through the 2nd stage, this minimizes power consumption.

Now all transistors in 1st and 2nd stage are scaled up such that first stage and 2nd stage have gm's that are the same as calculated in assignment 5. As the compensation capacitor and load capacitor are large, details of biasing and transistor sizing only affect the DC gain, near unity gain frequency, the values of both gm's is most important.

Both the gm's are then tuned by scaling each stage, to give better phase margin and bandwidth, based on simulated loop gain. If unity gain frequency is too low, gm1 has to be increased, while if 2nd pole is too low, gm2 has to be increased.